QP Code: 25901814

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION, APRIL 2026
Second Semester
Discipline Specific Core Course (DSC) –MG2DSCZGY100, MG2DSCZIM100
ENVIRONMENTAL BIOLOGY
(2024 ADMISSION ONWARDS)

Time: 1 hour 30 min **Max. Marks: 50**

Part A
Multiple Choice Questions
*Answer **all** questions*
*Each question carries **1 mark***

1. **The South American low grasslands are known as ……** K/CO1
   (a) Pampas

2. **……. is the author of the famous book 'Man and Nature'.** K/CO1
   (b) George Perkins Marsh

3. **Select the primary ecological role of a predator.** U/CO2
   (c) Controlling prey populations and maintaining balance

4. **Name an effect of air pollution.** K/CO3
   (a) Acid rain

5. **Identify the primary function of the National Biodiversity Authority under the National Biodiversity Strategy and Action Plan.** U/CO3
   (b) Regulating biological resource access

6. **Expand WWF.** K/CO3
   (d) World Wide Fund

7. **A rapid and intense flood that occurs within six hours of heavy rain:** K/CO4
   (d) Flash Flood

8. **The Environment Protection Act, 1986, empowers the government to create rules related to which of the following?** K/CO4
   (d) All of the above

9. **The primary challenge in implementing solid waste segregation:** U/C04
   (b) Public awareness and participation

10. **The organisation which plays a significant role in watershed development program in India.**
    (b) IWDP (The Integrated Watershed Development Programme) K/CO4

11. **Bhopal Gas Tragedy was caused by …….** K/CO4
    (a) MIC

12. **Kyoto Protocol was aimed at …….** K/CO4
    (a) Reduction of GHG emissions

(1 X 12= 12 Marks)

Part B

Short answer questions

*Answer any **6** questions*

*Each question carries **3 marks***

13. Discuss how human activities impact decomposers in an ecosystem. U/C01

Human activities including deforestation, mining, urbanisation, pollution due to pesticides and heavy metals etc destroy the habitat of decomposers (fungi, bacteria and invertebrates) which slows down the decomposition process and thereby nutrient cycling which ultimately affect ecosystem health. Excessive nutrient runoff (nitrogen and phosphorus) can change soil or water chemistry, reduce fungal diversity and alter the efficiency of decomposition. Changes in temperature and moisture levels can disrupt the specialized conditions needed by many decomposers to thrive. (**Liberal Valuation**)

14. Distinguish between food chain and food web with examples. U/CO1

A food chain is a linear sequence of organisms in an ecosystem, where each organism serves as a source of food (energy) for the next organism in the chain. It illustrates the flow of energy and nutrients through an ecosystem, starting with primary producers and moving up through consumers.

Example 1: **Grass → Grasshopper → Frog → Snake → Hawk** (terrestrial)

Example 2: **Phytoplankton → Zooplankton → Small Fish → Tuna → Shark** (aquatic).

A food web is a more complex and realistic representation of the feeding relationships in an ecosystem compared to a food chain. While a food chain shows a linear sequence, a food web illustrates how different food chains in an ecosystem are interconnected, showing the many pathways through which energy and nutrients flow.

Example: In a forest ecosystem where **plants** are eaten by **rabbits, squirrels, and insects**, which are then consumed by **foxes, snakes, and hawks**.

(*Any two distinguishing features with any one example for each, give full credit*)

15. Differentiate between absolute natality and ecological natality. U/CO2

Absolute Natality: The maximum number of new individuals an organism can produce under ideal conditions.

Ecological Natality: The increase in population size under real, environmental conditions.

16. Describe exponential growth curve. U/CO2

In exponential population growth, the population increases at a constant rate over time, assuming there are unlimited resources and no environmental limitations. The growth rate is directly proportional to the current population size resulting in a J-shaped curve when represents in a graph. This curve usually does not flatten out, though some models may show the growth rate slowing down as the population approaches the environment's carrying capacity

17. Examine the role of women in the Chipko Movement. U/CO3

Chipko Movement identified chiefly as a women movement, originated in the Garhwal Himalayan Mountain region of the Uttarakhand state in 1970s. Its aim was to prevent deforestation and conserve forests. Mira Behn and Goura Devi were the women leaders of the movement. Women act as the primary agents of change by pioneering the strategy of hugging trees (Chipko) to prevent deforestation. As they depended on forests for fodder, firewood, and water, local women led by figures like Gaura Devi led protests to protect their livelihood from commercial exploitation. The movement became a platform for rural women to emerge as community leaders, shifting from traditional roles to active political protestors (Women empowerment and social change).

18. List 3 types of human activities allowed in wildlife sanctuaries. K/CO3

- ***Private ownership rights***
- ***Timber harvesting***
- ***Collecting minor forest products***: Gathering honey, fruits, berries, and medicinal plants
- ***Regulated Eco-tourism***: Wildlife photography, safaris, and bird watching.
- ***Sustainable Resource Use***: Regulated grazing of livestock and, in some cases, limited, sustainable agriculture

(*Any three points, give full credit*)

19. India is a mega-diversity nation. Explain. U/CO3

Nearly 7-8% of the recorded species are present in India. 4 of the 34 globally identified biodiversity hotspots are there. India is a vast repository of traditional knowledge associated with biological resources. Over 91,200 animal species and 45,500 plant species have been documented in the ten biogeographic regions of the country. Along with species richness, India also possesses high rates of endemism. Wide range of ecosystems and habitats are present due to varied climatic, topographic and geological conditions. India has vast coastline and associated wetlands which holds wide varieties of rare and endangered species too. Nine biosphere reserves, six 'Ramsar sites' and five 'World Heritage Sites' are also present in India.

20. Interpret what a high Simpson's Index value reveals about a community. U/CO3

A high value in Simpson's Index of Diversity (1-D) reveals a community with high biodiversity, characterized by a large number of species and a relatively even distribution of individuals among those species. It indicates high species richness, high species evenness, ecosystem stability and resilience and no dominance of any species.

(3 X 6 = 18 Marks)

Part C

Short essay questions
*Answer any **4** questions*
*Each question carries **5 marks***

21. Discuss the role of bacteria in maintaining the nitrogen cycle. U/CO1

Bacteria are crucial for maintaining the environmental balance of nitrogen. They drive the transformation of nitrogen into different chemical forms through nitrogen fixation (*Rhizobium, Acetobacter*), nitrification (*Nitrosomonas, Nitrobacter*), ammonification, and denitrification (*Pseudomonas, Clostridium*). Without these microorganisms, nitrogen would accumulate in unusable forms (nitrites), leading to decreased soil fertility and limiting plant growth. They are also essential in wastewater treatment and agriculture, providing natural fertilization.

(*Brief explanation of nitrifying process, give full credit*)

22. Compare and differentiate keystone species and umbrella species U/CO2

Keystone Species: The species that have an extremely high impact on a particular ecosystem relative to its population. The removal of a keystone species from a habitat results in a dramatic change in the habitat. These species affect the structure and function of an ecosystem and the plant and animal species residing in that ecosystem. E.g., Sea otters, Beavers

Umbrella Species: The species which has a habitat that is important to the survival of many other animals. An umbrella species provide protection to a wide range of other species and their habitats. These species are often selected for conservation efforts because their survival inherently safeguards ecosystems and biodiversity. E.g., Giant Panda, Asian Elephant

23. Assess how the UN's Sustainable Development Goal 15 of the 2030 Agenda addresses biodiversity loss and ecosystem restoration. An/CO3

UN Sustainable Development Goal 15 (SDG 15) serves as a critical, actionable framework to "Life on Land" by aiming to protect, restore, and promote sustainable terrestrial ecosystems by 2030. The SDGs are structured around five pillars: People, Planet, Prosperity, Peace and Partnerships.

It addresses biodiversity loss by focusing on:

- Conserving, restoring and using terrestrial ecosystems in a sustainable manner
- Fighting against desertification
- Rehabilitating degraded lands and soils
- Conserving mountain ecosystems, including their biological diversity
- Taking measures to end poaching and trafficking of protected flora and fauna.

24. Examine how biodiversity conservation benefits human society and provide examples. A/CO3

Biodiversity conservation benefits human society in many direct and indirect ways. Biodiversity includes the variety of plants, animals, microorganisms, and ecosystems on Earth. Protecting it helps maintain ecological balance and supports human life, health, and economies. The benefits for humans by biodiversity includes:

- ***Provides food and medicines*** (Crops, Livestock, medicines derived from plants, fungi, and microorganisms)
- ***Maintains ecosystems services*** (Clean air and water, Soil fertility, Pollination, Flood control, Climate regulation)
- ***Supports economic development*** (by means of ecotourism, forest products, sustainable fishing etc.)
- ***Protects against natural disasters and climate change*** (Coral reefs protect coastlines, Forests reduce landslides and regulate rainfall, Diverse agricultural systems are better able to survive droughts and pests)
- ***Preserves cultural and recreational values***
- ***Ensures scientific and future benefits***

(*Points with brief explanation and example, give full credit*)

25. Explain how the Forest Conservation Act, 1980, protects forest areas from unauthorised use or destruction. U/CO4

The Forest (Conservation) Act, 1980 was enacted by the Government of India to protect forests from deforestation, illegal diversion, and non-forest activities. The Act places strict controls on the use of forest land and helps conserve biodiversity and ecological balance.

The main ways by which the Act protects Forest Areas are as follows:

- The State Government need mandatory approval from the Central Government for forest land use prevents uncontrolled clearing of forests for industries, mining, roads, dams, or commercial projects.
- Restricts non-forest activities like mining, commercial agriculture, industrial expansion etc.
- Prevents deforestation which helps to protect wildlife habitats, maintain rainfall and climate balance etc.
- Compensatory afforestation
- Penal action against violations
- Supports wildlife and biodiversity conservation

26. Explain the role of carbon trading in achieving national or global carbon reduction targets. U/CO4

Carbon trading is a system where governments set a limit (cap) on total carbon emissions. It plays an important role in helping countries and industries reduce greenhouse gas emissions and achieve climate goals. It is a market-based system that encourages companies to cut carbon emissions in a cost-effective way.

Carbon Trading helps achieve Carbon Reduction targets by:

- Encouraging industries to reduce emissions
- Making emission-reduction cost-effective
- Supporting international climate goals
- Promoting Renewable energy and green innovation

(*Brief explanation of relevant points, give full credit*)

(5 X 4 = 20 Marks)

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